

Campus Maintenance Intelligence

An Intelligent Campus Maintenance Management System

Project Proposal for UCS503P

Submitted to:

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1 Introduction

Campus Maintenance Intelligence is a smart maintenance management system designed for educational campuses. It manages complaints related to electrical systems, plumbing, AC, housekeeping, masonry, painting, fire systems and other facilities.

Traditional maintenance systems generally process complaints using first-come-first-served methods. This may result in critical problems being handled after less important complaints.

The proposed system uses complaint information, location, severity, affected users and historical maintenance data to assign suitable priorities.

The system follows five major stages:

Structure → Cluster → Prioritize → Predict → Optimize

This allows the system to move from simple complaint registration towards intelligent and proactive maintenance.

2 Problem Statement

Large campuses generate many maintenance complaints every day. Handling these complaints manually creates several problems:

- Critical complaints may not receive immediate attention.
- Multiple users may report the same problem.
- Recurring failures are difficult to identify.
- Historical maintenance data is not fully utilized.
- Technician and maintenance resources may be allocated inefficiently.

Therefore, a system is required that can automatically prioritize complaints, identify duplicates, predict possible failures and assist in maintenance planning.

3 Proposed Solution

The proposed system provides a centralized platform for campus maintenance.

When a complaint is submitted, the system:

1. Extracts important information from the complaint.
2. Identifies similar or duplicate complaints.
3. Assigns a priority based on severity, safety, location and affected users.
4. Stores the complaint as an incident or work order.
5. Uses historical data to identify possible future failures.
6. Helps optimize maintenance resources.

For example, an electrical short circuit in a hostel can receive a higher priority than a routine fan repair even if the fan complaint was submitted earlier.

4 Project Objectives

The main objectives are:

- Develop a centralized campus maintenance platform.
- Automatically prioritize maintenance complaints.
- Detect duplicate and related complaints.
- Identify recurring maintenance problems.
- Predict possible future failures.
- Improve technician and resource allocation.
- Provide a dashboard for maintenance staff.
- Compare intelligent prioritization with first-come-first-served processing.

5 System Overview

The system consists of the following major modules:

Module	Purpose
Complaint Management	Register and track maintenance complaints.
Priority Engine	Assign priority according to severity and campus context.
Duplicate Detection	Identify complaints referring to the same problem.
Incident Management	Combine related complaints into a single incident.
Predictive Maintenance	Identify assets that may fail in the future.
Optimization	Help allocate technicians and maintenance resources.
Dashboard	Display complaints, priorities and maintenance statistics.

Table 1: Major System Modules

6 Technology Stack

Component	Technology
Frontend	Next.js, React.js, Tailwind CSS
Backend	FastAPI, Python
Database	PostgreSQL
Machine Learning	XGBoost, Random Forest
NLP	Sentence Transformers
Similarity Search	FAISS
Data Processing	Pandas, NumPy
Optimization	OR-Tools
Version Control	Git, GitHub
Deployment	Docker

Table 2: Technology Stack

7 Dataset

The project will use complaint and historical maintenance data.

Complaint data will contain information such as:

- Complaint description
- Category and problem type
- Building and location
- Timestamp
- Priority
- Incident ID

Historical maintenance data will contain information such as building details, maintenance work orders, repair costs, maintenance priority and labor hours.

A synthetic benchmark dataset will initially be used for testing. It will represent campus locations such as hostels, blocks, lecture theatres, the CSED department and the library.

The dataset will contain maintenance categories including electrical, plumbing, AC, carpentry, masonry, painting, housekeeping and fire systems.

8 Development Methodology

The project will be developed using an Agile approach.

The major stages are:

1. Requirement analysis
2. Dataset preparation
3. Database and system design
4. Backend development
5. Frontend development
6. Machine learning development
7. System integration

8. Testing
9. Deployment

9 Expected Outcomes

The project is expected to provide:

- Faster identification of critical complaints.
- Reduction in duplicate complaint handling.
- Better maintenance prioritization.
- Identification of recurring problems.
- Early detection of possible failures.
- Better utilization of maintenance resources.
- A centralized maintenance dashboard.

10 Future Scope

The system can later be extended with:

- IoT sensors for real-time infrastructure monitoring.
- Image-based detection of infrastructure damage.
- Mobile application for maintenance workers.
- Automated technician assignment.
- Advanced failure prediction.
- Integration with the actual university maintenance system.

11 Conclusion

Campus Maintenance Intelligence aims to transform traditional campus maintenance into a more intelligent and data-driven process.

The system combines complaint management, priority assignment, duplicate detection, predictive maintenance and resource optimization in a single platform.

By using historical and synthetic maintenance data, the system can be developed and evaluated before deployment in a real campus environment.

Campus Maintenance Intelligence

Report. Prioritize. Predict. Optimize.